

PLANNING FOR PREGNANCY, STRESSED ABOUT FOOD: FOOD INSECURITY AMONG WOMEN IN THE PRECONCEPTION PERIOD

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INTRODUCTION

Food insecurity, the inconsistent access and availability of sufficient safe, affordable, and nutritious food, is a growing public health issue affecting more than 13% of Australian households [1]. Research from the state of Victoria further suggests that two-thirds of food insecure Victorians are between the ages of 18-44 years [2], underlining the disproportionate burden to those of reproductive age. The preconception period is increasingly recognised as a crucial stage in reproductive healthcare to shape pregnancy outcomes [3]; however, evidence of how food insecurity affects women during this period remains limited. This study aimed to identify food insecurity prevalence among women in the preconception period living in Victoria, Australia; assess sociodemographic characteristics of food insecure women; and explore the association of food insecurity on pregnancy planning.

METHODS

A cross-sectional online survey was conducted (April 2025 - March 2026) with women aged 18 to 44 years, residing in Victoria, Australia, who were proficient in English, and intending to have a future pregnancy. Food security status was assessed using the USDA Household Food Security Survey Module (10- and 18-item depending on parental status), alongside survey items to assess sociodemographic and pregnancy planning factors. Descriptive statistics summarised sample characteristics. Food security status was reported as binary food secure or food insecure categories, and by food security categories (high, marginal, low, and very low food security). Poisson regression with robust standard errors was used to examine associations between sociodemographic variables and food security prevalence, with prevalence ratios (PR) and 95% confidence intervals (CIs) reported. Covariates were adjusted for with age, gender, born in Australia, educational attainment, and household composition.

RESULTS AND DISCUSSION

Survey data was collected from 226 women (mean age 30.3 ± 5.7 years), with almost half identified as living in a food-insecure household (43.8%), and more than half once applied to the population (56.1%), comprising marginal (11.5%), low (18.5%), and very low food security (26.1%). Food insecurity was more prevalent

among households with women experiencing unemployment, lower educational attainment, lower household income, and living in temporary accommodation. Women living in food-insecure households were also more likely to report that pregnancy timing is influenced by food-related factors, including food costs (PR = 2.21, 95% CI: 1.57 – 3.13, $p < 0.001$), inadequate and inconsistent food access (PR = 1.57, 95% CI: 1.16 – 2.12, $p = 0.004$), stress/worry about food cost (PR = 1.80, 95% CI: 1.18 – 2.73, $p = 0.006$), anxiety/stress about having enough food during pregnancy (PR = 1.87, 95% CI: 1.38 – 2.53, $p < 0.001$), and anxiety/stress about pregnancy complications from adequate or insufficient food (PR = 1.64, 95% CI: 1.17 – 2.32, $p = 0.005$).

CONCLUSIONS

The findings of this study suggest food insecurity affects 56% of women in the preconception period living in Victoria, Australia. Women living in food insecure households were associated with socioeconomic disadvantage and food-related factors associated with pregnancy planning and timing, and mental health. Further research is needed to better understand food insecurity in the preconception period regarding what actions should be taken to target prevention and management of food insecurity prior to pregnancy, ensuring optimal health of women and their babies across the perinatal period.

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